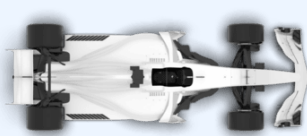
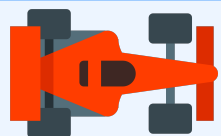




F1 Historical Dataset (Kaggle):
<https://www.kaggle.com/datasets/rohanrao/formula-1-world-championship-1950-2020>



F1 Historical Weather Dataset (Kaggle):
<https://www.kaggle.com/datasets/mariyakostyrya/formula-1-weather-info-1950-2024>



Data is manually loaded/batched into an Snowflake Warehouse (X-SMALL) which contains an F1 Database. You can use AWS, GCP Pub/Sub, or Azure storage buckets and create an auto-ingest pipeline. If you want to use live race data, you can use Kafka.

I used a free trial Snowflake account as I wanted to get familiar with the interface (coming from databricks). You can use databricks or use local databases (DuckDB, other alternatives)



Data Modeling (Star schema 1 - GLOBAL_CONSTRUCTOR_STANDINGS - Racing Team total Stats)

Note: SQL operations peformed: CTEs, JOINS, AGGREGATE FUNCTIONS (MAX, AVG, etc.), and general data formatting. DIMENSION TABLES in Blue, FACT TABLES in Black.

CONSTRUCTORS (VARCHAR)
- CONSTRUCTORID
- CONSTRUCTORREF
- CON_NAME
- NATIONALITY
- REF_URL

CONSTRUCTOR_RESULTS
- CONSTRUCTORID (NUMBER)
- CONSTRUCTORRESULTSID (NUMBER)
- POINTS (NUMBER)
- RACEID (NUMBER)
- STATUS (VARCHAR)

GLOBAL_CONSTRUCTOR_STANDINGS
- CONSTRUCTOR_ID
- BRAND_NAME
- RACE_ID
- CONSTRUCTOR_POINTS
- CON_STANDING_POINTS
- CONSTRUCTOR_POSITION
- CONSTRUCTOR_WINS

CONSTRUCTOR_STANDINGS
- CONSTRUCTORID (NUMBER)
- CONSTRUCTORSTANDINGSID (NUMBER)
- POINTS (NUMBER)
- POSITIONTEXT (VARCHAR)
- RACEID (NUMBER)
- WINS (NUMBER)

Data Modeling (Star schema 2 - GLOBAL STANDINGS STATS - Individual Driver stats)

DRIVERS (VARCHAR)
- DRIVER ID
- DRIVERREF
- DRIVER_CODE
- DRIVER_DATEOFBIRTH
- DRIVER_NATIONALITY
- DRIVER_NUMBER
- DRIVER_REF_URL
- DRIVER_SURNAME

DRIVER_STANDINGS
- DRIVERID (NUMBER)
- POINTS (NUMBER)
- POSITION (NUMBER)
- RACEID (NUMBER)
- WINS (NUMBER)

GLOBAL_STANDINGS
- DRIVER_ID
- DRIVER_REF
- DRIVER_NAME
- RACE_NAME
- RACE_ID
- RACE_YEAR
- STANDING_POINTS
- STANDING_POSITION
- STANDING_WINS
- RACE_FINISH_POSITION
- RACE_POINTS
- GRID_POSITION
- FASTEST_LAP
- FASTEST_LAP_TIME
- FASTEST_LAP_SPEED

RACES
- NAME (VARCHAR)
- YEAR (NUMBER)
- RACEID (NUMBER)
- CIRCUITID (NUMBER)
- ROUND (NUMBER)
- DATE (DATE)
- Other columns present (First place lap time, Second place lap time, etc. NOT USED/REFERENCED)

RESULTS
- POSITION (NUMBER)
- POINTS (NUMBER)
- GRID (NUMBER)
- FASTESTLAP (NUMBER)
- FASTESTLAPTIME (VARCHAR)
- FASTESTLAPSPEED (NUMBER)
- DRIVERID (VARCHAR)
- RACEID (VARCHAR)

Note: Standing Position is the F1 driver ranking AFTER the race has concluded. Race Position refers to the positional finish in a single Grand Prix (2-3 hour race)

Data Modeling (Star schema 3 - FACT_PIT_STOP_DATA - Granular Pit Stop Data)

WEATHER_FEATURES
- DATETIME (TIMESTAMP)
- NAME (VARCHAR)
- PRECIPITATION (NUMBER)
- ROUND (NUMBER)
- TEMPERATURE (NUMBER, in Celsius)
- WINDSPEED (NUMBER)

PIT_STOPS
- DRIVERID (NUMBER)
- DURATION (VARCHAR)
- LAP (NUMBER)
- MILLISECONDS (NUMBER)
- RACEID (NUMBER)
- STOP (NUMBER)
- TIME (TIME)

FACT_PIT_STOP_DATA
- RACE_ID
- YEAR
- ROUND
- RACE_NAME
- RACE_DATE
- DRIVER_ID
- CONSTRUCTOR_ID
- NUM_PIT_STOPS
- FASTEST_PIT_STOP_MS
- SLOWEST_PIT_STOP_MS
- AVG_PIT_STOP_MS
- SPRINT_GRID
- SPRINT_POSITION
- SPRINT_POSITION_TEXT
- SPRINT_POSITION_ORDER
- SPRINT_POINTS
- SPRINT_LAPS
- SPRINT_TIME
- SPRINT_MILLISECONDS
- SPRINT_FASTEST_LAP
- SPRINT_FASTEST_LAP_TIME
- SPRINT_STATUS_ID
- AVG_TEMPRATURE
- MIN_TEMPERATURE
- MAX_TEMPERATURE
- TOTAL_PRECIPITATION
- AVG_WINDSPEED
- MAX_WINDSPEED

SPRINT_RESULTS
- GRID (NUMBER)
- POSITION (NUMBER)
- POSITIONTEXT (VARCHAR)
- POSITIONORDER (NUMBER)
- POINTS (NUMBER)
- LAPS (NUMBER)
- TIME (VARCHAR)
- MILLISECONDS (NUMBER)
- FASTESTLAP (NUMBER)
- FASTESTLAPTIME (VARCHAR)
- STATUSID (VARCHAR)
- DRIVERID (VARCHAR)
- CONSTRUCTORID (NUMBER)

RACES
- NAME (VARCHAR)
- YEAR (NUMBER)
- RACEID (NUMBER)
- CIRCUITID (NUMBER)
- ROUND (NUMBER)
- DATE (DATE)
- Other columns present (First place lap time, Second place lap time, etc. NOT USED/REFERENCED)



Dashboard of Pit Stop Fact table Metrics
Google Data Studio + Snowflake BI Connector

